Chapter 2
Mountain Rain Forests in Southern Ecuador as a Hotspot of Biodiversity – Limited Knowledge and Diverging Patterns

G. Brehm(✉), J. Homeier, K. Fiedler, I. Kottke, J. Illig, N.M. Nöske, F.A. Werner, and S.-W. Breckle

2.1 Introduction: Why Do We Need Biodiversity Inventories?

Highly complex ecosystems such as the tropical mountain rain forest in southern Ecuador probably harbor tens of thousands of species that interact with each other. It is impossible to understand an ecosystem without knowing the composition of its community. Such knowledge cannot be achieved without the examination of all major groups of animals, fungi, plants, and bacteria. For example, insects such as leaf beetles, ants, or hymenopteran and dipteran parasitoids have a high impact on forest ecosystems (Moutino et al. 2005; Soler et al. 2005), but have not been studied at the RBSF so far. The question of how many species there are on earth is still unresolved. Estimates range from four to 30 million species (e.g. Novotny et al. 2002). Ultimately, only counting and naming species can answer this question.

A hot debate about the methodological approaches, i.e. the usefulness and efficiency of 'traditional' taxonomy versus DNA barcoding approaches is still going on (e.g. Meyer and Paulay 2005; Will et al. 2005). Barcoding techniques provide useful sets of new characters for species descriptions and phylogenies, especially in cryptic species or taxa otherwise poor in morphological characters (e.g. Hajibabaei et al. 2006). However, species definition based on DNA sequence data can be problematic. For example, ribosomal genes can show intraspecific variation in the multinucleate Glomeromycota. The sequence types obtained from the vegetative stage can rarely be related precisely to either morphological or biological species (Sanders 2004). In our opinion, the only strategic way forward can reside on a synthesis of 'classic' and 'modern' approaches with regard to sampling campaigns, application of up-to-date information technology (Godfray 2005), and thorough taxonomic and systematic work. This might include in vitro cultivation, in the case of fungi and bacteria.

Traditional descriptive disciplines such as taxonomy, systematics and natural history provide names, phylogenies, and life history data as a service for other fields of science and their applications. However, these disciplines have suffered a great loss of capacity in past decades (e.g. Breckle 2002; Gotelli 2004). Many species identifications from the RBSF are not the primary result of research efforts in

E. Beck et al. (eds.), *Gradients in a Tropical Mountain Ecosystem of Ecuador.* *Ecological Studies 198.*

taxonomy and systematics. Rather, the recently published species checklists (Liede-Schumann and Breckle, 2008) are essentially a by-product of ecological and physiological research. This explains why many lists are still far from being complete (see below) and why only a few species have been newly described from the area thus far. Many (probably many thousands) of species in the RBSF alone are new to science and await description, but specialists are lacking for many taxa. Only some vertebrate groups are well known: All currently recorded bird and bat species are described (Matt 2001; Muchhala et al. 2005; Paulsch 2008). In other groups, however, the proportion of described species drops: from bryophytes (98%; Gradstein et al. 2008), trees (>5 cm dbh, 90%), lichens (85%; Nöske et al. 2008), to geometrid moths (63%; Brehm et al. 2005), oribatid mites (60%; Illig et al. 2008; Niedbala and Illig 2007) to fungi (5%; Haug et al. 2004; Setaro et al. 2006a;, Suarez et al. 2006; Kottke et al. 2007).

2.2 Inventory Coverage in the RBSF

After eight years of research in southern Ecuador, our knowledge about certain taxa is excellent. The area is one of the most thoroughly investigated neotropical montane forests (see the contributions to this volume). Geometrid and pyraloid moths, and oribatid mites or bush crickets (Tettigoniidae) have never been studied quantitatively in other Andean forests before (Süßenbach 2003; Brehm et al. 2005; Braun 2008; Illig et al. 2008). None of the mycorrhizal fungi has previously been known from the northern Andean forests (Haug et al. 2004, 2005; Setaro et al. 2006a; Suarez et al. 2006; Kottke et al. 2007). However, there is little reason to rest on the laurels. Our knowledge about species richness in the area remains biased and incomplete (Fig. 2.1).

The Bacteria or Archaea have not been investigated in the study area. So far, the only group of single-cell organisms studied are the Testacea (Krashevska 2008; Table 2.1). Due to their dominant role in material and energy flows in ecosystems, plants have received relatively much attention, and groups such as vascular and non-vascular cryptogams have been inventoried across the whole elevational range (see Chapter 10.1). However, botanical collections have focused on the existing trail system that is biased towards ridge sites. Bryophytes and ferns have been collected systematically across the whole altitudinal range (e.g. Kürschner and Parolly 2004a; see Chapter 19; Table 2.1).

Lichens have been sampled intensively at least in the lowermost parts (up to 2100 m) of the area (e.g. Nöske 2005). Moreover, life forms such as trees (Homeier 2004), climbers (S. Matezki, personal communication) and epiphytes (Werner et al. 2005) have been treated in extensive ecological studies, whereas terrestrial shrubs and especially herbs still remain poorly known. Concerning the large group of fungi, only mycorrhiza forming groups were investigated. The molecular diversity of the Glomeromycota forming mycorrhizas with trees and two basal groups of Basidiomycota (Sebacinales, Tulasnellales) forming mycorrhizas with orchids, ericads,